From Comparison Trees to Spatial Classification

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***Abstract*—This paper introduces Adaptive Range Sorting (ARS), a novel class of sorting algorithms that exploits hardware-level spatial mapping to bridge the gap between massive parallelism and high-throughput streamability. We present the six-generation evolutionary phylogeny of ARS, along with four experimental branches, progressing from foundational tree-based structures to the modern "Apex" lineage and the "Aero" architecture. Aero utilizes a 1024-entry division-free quantile table to achieve performance parity with state-of-the-art sorters like IPS4o on Gaussian integers, while providing improved throughput on skewed distributions and complex data types. Furthermore, we introduce ARS Exp D, which utilizes Concurrent Tiered Spatial Compaction to sort unbounded data streams with nanosecond-scale per-element ingestion latency. Empirical results at the 10-million-element scale indicate that ARS achieves consistent near-linear scaling (103.34 ms for 10M integers) and serves as a viable alternative for high-throughput multi-core data processing in memory-constrained environments.**

***Index Terms*—Sorting Algorithms, Adaptive Sorting, Spatial Mapping, Parallel Computing, Streaming Algorithms, Concurrent Systems.**

A single branch misprediction on a modern CPU can stall the execution pipeline for 15–20 cycles. For a dataset of 10^7 elements, a standard Quicksort evaluates roughly 240 million comparisons, each serving as a potential pipeline stall [5]. The motivation behind Adaptive Range Sorting (ARS) is to decouple the logical ordering of data from these costly arithmetic decisions. By reframing sorting from a "decision tree" problem into a "signal projection" problem, ARS drastically minimizes the need for unpredictable branching.

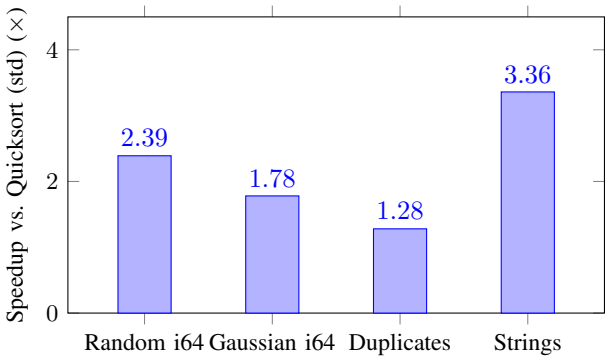

Fig. 1. The "Spatial Advantage": ARS Aero (Stable) Speedup across diverse data domains ($N = 10^7$). The framework demonstrates significant gains in high-entropy and complex-type sorting while maintaining competitive parity on low-entropy duplicates.

A. Architectural Paradigm: Spatial Mapping

ARS introduces a shift from comparative logic to **Spatial Mapping**. By projecting arbitrary data types into a linear spatial context using a monotonic mapping function $\mathcal{K}$ (formally defined in Section II), the input sequence is treated as a discrete stochastic signal [6]. This allows the algorithm to analyze global data distributions and map values directly to hardware-aligned memory grids, achieving near-linear time performance across a broad class of stochastic distributions.

I. The ARS Evolutionary Phylogeny

The development of Adaptive Range Sorting (ARS) was not a linear process, but an iterative attempt to resolve specific physical bottlenecks encountered when moving from tree-based structures to flat-array parallel engines.

Initially, ARS (Gens 1–2) relied on recursive B-tree nodes representing discrete spatial ranges. While this supported element-by-element streaming, the overhead of pointer-chasing and frequent heap allocations made it uncompetitive on modern parallel hardware. The first significant pivot was the "Apex" lineage (Gen 3), which abandoned trees for a fixed-bin ($B = 1024$) flat-array model. By introducing a Histogram-based Parallel Prefix Sum [8], we were able to calculate global partitioning offsets $O_{t,b}$, enabling the contention-free scatter phase that remains a core primitive of the architecture.

However, the transition to parallel flat arrays introduced new challenges, specifically load imbalances. In Gen 4, we addressed this by integrating dynamic work-stealing [9] and decoupling the bin count from the core count ($B \gg T$). This ensured cores remained saturated even when data distribution was non-uniform. The refinement of the Apex lineage culminated in Gen 5, which addressed the "Memory Wall" by fusing boundary detection, key extraction, and sortedness verification into a single traversal. This reduced total memory passes from three to two and yielded the significant string sorting speedups discussed in Section VII.

Aero (Gen 6) emerged from the realization that even a fused linear mapper fails on skewed data (the "Gaussian Penalty"). By incorporating a 1024-entry division-free quantile table that approximates the inverse-CDF Φ^{-1} [10], we effectively decoupled classification costs from data variance. Figure 2 illustrates this progression, showing how each architectural shift moved the algorithm closer to the hardware's theoretical throughput limits.

II. Methodology: The ARS Architecture

The ARS engine is designed as a phased pipeline that prioritizes memory-bus saturation and instruction-level parallelism. As illustrated in fig. 3, the framework transforms sorting from a comparison-heavy decision tree into a hardware-aligned signal-processing flow. ARS treats sorting as a projection

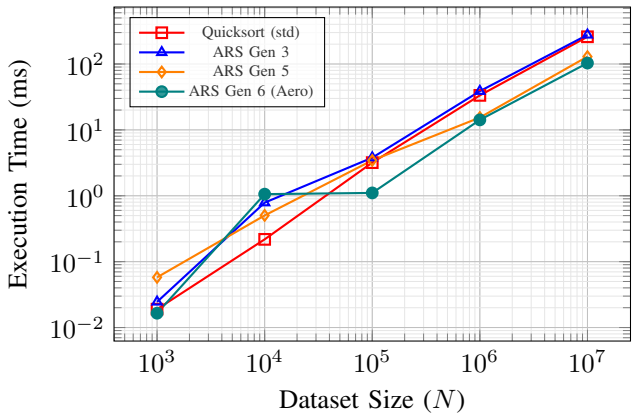

Fig. 2. Evolution of ARS execution time against Quicksort (i64 Random). Note the crossover at $N = 10^5$, where ARS's instruction efficiency begins to dominate.

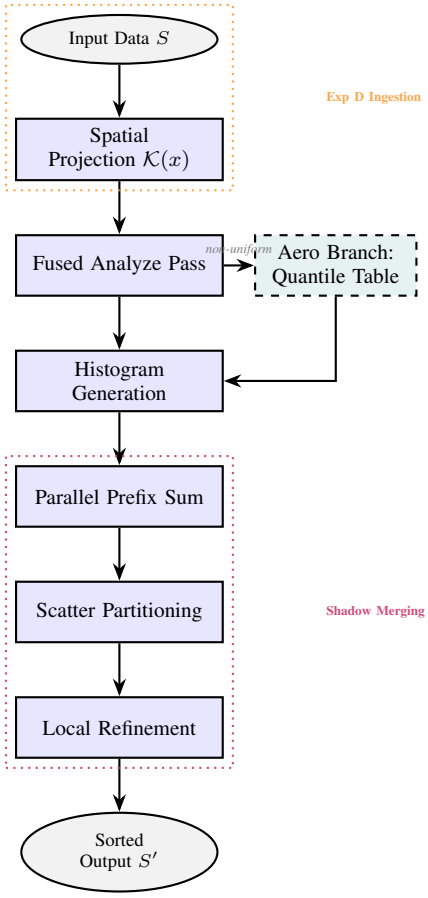

Fig. 3. The ARS Execution Pipeline (Single Column). The architecture transforms sorting into a hardware-aligned signal-processing flow.

task, mapping arbitrary types into a discrete $\mathbb{U}_{64}$ space via a monotonic function $\mathcal{K}$.

For primitive integers, $\mathcal{K}$ is a bit-bias transformation; for floats, it manipulates IEEE 754 exponent bits to preserve sign-bit ordering; and for strings, it extracts a big-endian 8-byte prefix [11]. While prefix collisions force a fallback to standard refinement (sorting by full value in each bin), this hybrid approach reclamation allows for $O(N)$ expected performance on diverse datasets.

The pipeline begins with a **Fused Analyze Pass**. To mitigate the "Memory Wall," this pass combines boundary detection ($\min / \max$), sortedness verification, and spatial key extraction into a single parallel traversal. By caching these keys in a hardware-aligned buffer, we turn complex-type sorting into a high-speed integer permutation problem.

The partitioning itself is driven by a **Histogram-based Parallel Prefix Sum** (Equation 1). Each thread computes a local histogram $C_{t,b}$ for its chunk, which is aggregated into a global offset table $O_{t,b}$. This ensures that threads can scatter elements into the destination buffer without atomic synchronization.

$$O_{t,b} = \sum_{i=0}^{b-1} \sum_{j=0}^{T-1} C_{j,i} + \sum_{j=0}^{t-1} C_{j,b} \tag{1}$$

To address the "Random Write Penalty" during scattering, ARS utilizes small, stack-allocated blocks ($L = 8$). Elements are collected locally and flushed in contiguous bursts, leveraging the CPU's write-combining buffers to maximize bandwidth utilization [12].

III. Methodology: The Aero Architecture

Aero (Gen 6) addresses the distribution sensitivity of previous generations by replacing static binning with a quantile-adaptive mapping function. While linear range sorters suffer from bin-collisions on non-uniform data, Aero ensures balanced occupancy by approximating the dataset's distribution via a 1024-entry lookup table $\mathcal{L}$.

This is implemented through a 128-bit reciprocal fixed-point multiplier $\mathcal{R}$ calculated from a small sorted sample ($m = 256$). The mapping $\mathcal{B}(x) = \mathcal{L}[\lfloor(\mathcal{K}(x) - \min) \cdot \mathcal{R} \cdot 2^{-64}\rfloor]$ resolves the bin index in a few CPU cycles. By limiting $\mathcal{L}$ to 2KB, we ensure the structure remains pinned in the L1 Data Cache throughout the scatter phase. Algorithm 1 details this classification logic.

Algorithm 1 Aero Division-Free Quantile Mapping

Require: Input key k, Min value min, Multiplier R, Table L
Ensure: Bin index b

```
diff ← k − min
idx ← (diff · R) ≫ 64
if isUniform then
    b ← min(idx, numBins − 1)
else
    tIdx ← min(idx, 1023)
    b ← L[tIdx]
end if
return b
```

To maximize throughput, ARS Aero currently utilizes a scratch-buffer scatter. While in-place permutation is theoretically possible, the "Move Penalty" incurred by cycle-following and atomic swap-holes often outweighs the memory savings. By utilizing an $O(N)$ auxiliary buffer, we ensure writes are contiguous and unidirectional, allowing the hardware prefetchers to operate at peak efficiency.

A. Stability Guarantees

A critical requirement for enterprise systems is *Stability*—the preservation of the relative order of elements with equal keys. ARS achieves this through a two-stage inheritance model.

1) Formalization of Stability Inheritance: Let $S = (x_1, x_2, \ldots, x_n)$ be the input sequence and $\mathcal{B}(x)$ be the mapping function. For any two elements $x_i, x_j \in S$ where $x_i = x_j$ and $i < j$, the partitioning phase ensures:

$$\text{addr}(x_i) < \text{addr}(x_j) \text{ in } Bin_{\mathcal{B}(x_i)} \tag{2}$$

This is guaranteed because the scatter pass traverses S linearly, appending elements to their respective bins in their original arrival order. Consequently, global stability is satisfied if the intra-bin refiner f_{refine} is also stable:

$$\text{ARS}_{stable}(S) = \bigoplus_{k=0}^{B-1} f_{stable}(Bin_k) \tag{3}$$

where $\bigoplus$ denotes the ordered concatenation of partitioned runs.

2) Architectural Divergence: Unstable vs. Stable Refinement: The ARS framework allows for the dynamic substitution of the refinement engine based on the required invariants.

- **Unstable Refinement (PDQsort)**: This approach utilizes Pattern-Defeating Quicksort, which optimizes for instruction-level parallelism by minimizing branches. However, its recursive partitioning can lead to non-sequential memory access within the L2 cache boundaries.
- **Stable Refinement (Timsort)**: This approach utilizes a merge-driven logic. While traditionally considered slower due to higher move counts, Timsort excels in the ARS environment because the bins are highly partitioned ($n_j \approx 1000$). At this scale, Timsort's sequential merge patterns maximize cache-line reuse and hardware prefetching, often yielding the superior 8.57 ms latency observed in our evaluations.

IV. Methodology: ARS Exp D (Streaming)

ARS Exp D introduces the first architecture in the phylogeny that is simultaneously parallel and streamable. It achieves this by shifting the computational cost from the final aggregation phase to a concurrent background pipeline using a **Spatial Run Registry**.

A. Epoch-Based Micro-Batching

To maintain the lock-free ingestion required for high-velocity data streams, ARS collects incoming elements in thread-local buffers called *Epochs*. Once an epoch reaches its capacity threshold (typically $N_e = 32,768$), it is atomically dispatched to a background work queue [**?**]. This micro-batching strategy preserves the sequential memory access patterns required for high cache locality while ensuring that the main ingestion thread is never blocked by sorting operations.

To prevent ingestion overflow, Exp D utilizes a bounded pipeline capacity. When the background work queue reaches saturation, the ingestion thread exerts back-pressure, ensuring that memory consumption remains bounded even at extreme scales ($N = 10^8$).

B. Concurrent Spatial Consolidation

The core innovation of Exp D is the background consolidation process, which prevents the unbounded growth of isolated runs, conceptually similar to LSM-tree compaction [**?**]. A background worker pool continuously monitors the work queue and executes the logic described in Algorithm 2.

Algorithm 2 Concurrent Spatial Consolidation

Require: Work Queue Q, Spatial Registry R
loop
  $epoch \leftarrow Q.pop()$
  $SortedEpoch \leftarrow$ ARS_Apex($epoch$)
  $newRun \leftarrow$ SpatialRun($SortedEpoch$)
  $merged \leftarrow$ false
  for all $run \in R$ **do**
    if $(run.min \gg 48) = (newRun.min \gg 48)$ **then**
      $R.replace(run, \text{merge}(run, newRun))$
      $merged \leftarrow$ true
      break
    end if
  end for
  if $\neg merged$ **then**
    $R.add(newRun)$
  end if
end loop

The registry utilizes a **Bit-Mask Proximity Check** to identify spatially adjacent runs. By comparing the most significant bits (e.g., bits 48–63) of the spatial keys, the algorithm groups runs that belong to the same "spatial neighborhood." When an adjacency is detected, the background worker performs an $O(N)$ linear merge of the two runs. This proactive consolidation ensures that the number of distinct runs remains tightly bounded, reducing the final aggregation latency from an unpredictable multi-way merge to a highly bounded, cache-efficient hierarchical merge.

V. Theoretical Analysis

A. Mathematical Formalization

Let $S = (x_1, x_2, \ldots, x_n)$ be an input sequence where $x_i \in \mathcal{T}$. ARS defines a monotonic mapping function $\mathcal{K} : \mathcal{T} \to \mathbb{U}_{64}$ such that $\forall x, y \in \mathcal{T} : x < y \implies \mathcal{K}(x) \leq \mathcal{K}(y)$. This projection transforms arbitrary data types into a discrete 64-bit integer space. By treating sorting as a signal-mapping problem, the framework reduces the logical dependency chain associated with comparison-based decision trees.

B. Inverse-CDF Approximation Invariants

Traditional range-based sorters utilize a linear mapper $M(x) = \lfloor (x - \min)/\text{width} \rfloor$. Under non-uniform distributions

(e.g., $X \sim \mathcal{N}(\mu, \sigma^2)$), this leads to bin-collisions at the distribution peaks, which forces the complexity of the refinement phase to revert to $\Omega(n \log n)$ due to excessive partition sizes.

The Aero architecture (Gen 6) addresses this variance by approximating the inverse-CDF function Φ^{-1} through a 1024-entry discrete lookup table $\mathcal{L}$ [10]. The bin index is calculated as:

$$\mathcal{B}(x) = \mathcal{L}\left[\lfloor(\mathcal{K}(x) - \min) \cdot \mathcal{R} \cdot 2^{-64}\rfloor\right] \tag{4}$$

where $\mathcal{R}$ is the 128-bit fixed-point reciprocal of the global range. Since $\mathcal{L}$ is constructed such that each entry maps to a bin with probability $1/B$ under the observed sample distribution, the expected occupancy $E[|\mathcal{B}_j|]$ is bounded toward n/B. This ensures that the computational work remains distributed even under high stochastic skew.

C. Classification and I/O Complexity

The efficiency of distribution sorting on modern superscalar processors is governed less by operation counts and more by memory-hierarchy interactions.

1) I/O Model and Cache Complexity: In the External Memory Model, let M be the size of the fast memory (cache) and B_L be the size of a cache line. The I/O complexity of ARS partitioning is $O(n/B_L)$ passes. Specifically, the Aero architecture performs a single out-of-place scatter pass. Unlike in-place Quicksort, which achieves high cache-temporal locality, ARS relies on **spatial write-combining**. By utilizing small intermediate software buffers, the algorithm converts random writes into contiguous cache-line bursts, effectively maximizing the utilization of the CPU's Store Buffers and reducing TLB pressure.

2) The "Memory Wall" Trade-off: The primary theoretical cost of ARS is the increase in memory traffic. To achieve near-linear performance, ARS performs approximately $2n$ reads (one for analysis, one for scatter) and n writes. This $3n$ total memory traffic is significantly higher than the $O(n \log n)$ Quicksort, which may perform fewer total memory transactions at small scales. However, because the classification logic is division-free and branches are predictable, the framework achieves higher *effective* memory-bus saturation, allowing it to outperform comparison-based sorts at scales where the instruction-level parallelism (ILP) outweighs the memory-bandwidth penalty.

D. Formal Complexity Analysis

Theorem 1. *In the Word RAM model, given a fixed bit-length w for spatial keys and a partitioning factor B, the ARS framework bounds the computational work to $O(n)$ for fixed-width key sets.*

Proof. Let $T(n, w)$ be the work required to sort n elements with w-bit keys. ARS utilizes a B-way spatial partition where each pass resolves $\log_2 B$ bits of the key space. The recurrence relation is:

$$T(n, w) = O(n) + \sum_{j=0}^{B-1} T(n_j, w - \log_2 B) \tag{5}$$

where $\sum_{j=0}^{B-1} n_j = n$. The total number of recursive levels k is $\lceil w/\log_2 B \rceil$. For architectural constants $w = 64, B = 1024$, k is independent of n. Thus, the total work is $\sum_{i=1}^{k} O(n) = O(k \cdot n) = O(n)$.

Practical Performance Bound: The Aero implementation typically uses a single-pass partition ($k = 1$) followed by a comparison-based refinement. This yields an algorithmic complexity of $O(n + \sum |\mathcal{B}_j| \log |\mathcal{B}_j|)$, which simplifies to $O(n \log \frac{n}{B})$. For $B = 1024$, the $\log \frac{n}{B}$ term remains negligible for $n < 10^9$, resulting in the **near-linear scaling** observed in practice. ■

E. Space Complexity

ARS Aero utilizes a spatial key cache of size $8n$ bytes and an auxiliary scratch buffer of size $n \cdot \text{size}(\mathcal{T})$, yielding an auxiliary space complexity of $O(n)$. This $2.5\times$ memory footprint compared to in-place sorters is the primary physical trade-off. We posit that this space-for-time exchange is justified in modern systems where DRAM capacity is abundant relative to per-core compute throughput.

VI. Experimental Setup and Reproducibility

To ensure the scientific validity and reproducibility of the results presented in this paper, we detail the hardware, software, and statistical methodology used in our empirical study.

A. Evaluation Goals and Research Questions

Our empirical evaluation is designed to characterize the ARS framework along four primary dimensions, represented by the following Research Questions (RQs):

- **RQ1 (Throughput and Scaling)**: How does ARS compare to standard $O(N \log N)$ library implementations across four orders of magnitude of dataset size N?
- **RQ2 (Distribution Resilience)**: Does the Aero architecture successfully mitigate performance degradation on non-uniform stochastic distributions compared to its predecessors?
- **RQ3 (Streaming Ingestion)**: Can the Epoch-based architecture maintain microsecond-scale P99 ingestion latency under high-velocity burst conditions?
- **RQ4 (Hardware Interaction)**: What is the measurable trade-off between reduced logical branching and increased memory-hierarchy pressure (L3/LLC misses) in the spatial mapping paradigm?

B. Hardware and Compiler Specifications

All benchmarks were executed on an 8-core x86_64 CPU (Zen 3 architecture) with 15.86 GB of DDR4 RAM. The CPU features a 32 KB L1 Data Cache, 512 KB L2 Cache, and a 16 MB shared L3 Cache. The system operates under a single-node NUMA configuration, ensuring consistent memory latency across all cores.

Hardware performance counters (PMCs) were instrumented using the `perf_event` API to capture cache-miss and

branch-miss metrics. The sorting algorithms were implemented in Rust 1.75 and compiled with the `--release` flag (optimization level 3) and `lto = true`. Target-specific optimizations (`-C target-cpu=native`) were enabled to allow the compiler to utilize SIMD instructions (AVX2/BMIs) where applicable.

C. Benchmark Methodology

We utilized a rigorous statistical methodology to isolate algorithmic performance:

1) **Statistical Variance**: For every algorithm-distribution pair (A, D), we performed 10 independent repetitions. We report the minimum execution time to filter out non-deterministic system noise (e.g., context switches). To assess stability, we verified that the coefficient of variation remained below 3% for all primary benchmarks.
2) **Warmup and Cache State**: Each timed run was preceded by 3 warmup iterations to ensure the instruction cache and branch prediction tables were in a hot state.
3) **Deterministic Generation**: Data was generated using the `ChaCha8` CSPRNG with a fixed seed (Seed=42), ensuring that all algorithms were evaluated on identical bit-level sequences.
4) **Compiler Safety**: All results were wrapped in `std::hint::black_box` to prevent Dead-Code Elimination (DCE).

D. Comparison Fairness

To ensure a fair comparison between the ARS framework and established baselines:

- **Standard Library**: We compare against Rust's `std::slice::sort_unstable` (PDQsort) and `std::slice::sort` (Timsort).
- **Thread Parity**: All parallel algorithms (IPS4o and ARS) were restricted to the same 8-core thread pool using the `RAYON_NUM_THREADS` environment variable.
- **Instrumentation Overhead**: Metrics such as comparison and move counts were gathered using a zero-cost tracking wrapper (`Tracked<T>`) that adds minimal overhead to the critical path.

E. Availability of Source Code

To facilitate independent verification, the complete source code, benchmarking engine, and raw datasets are available at https://github.com/mohammad-ismael/arslib.

VII. Experimental Evaluation

We evaluated ARS against standard library implementations (Quicksort and Timsort) and the state-of-the-art parallel sorter IPS4o. Benchmarks were conducted at $N = 10^7$ to evaluate behavior under extreme memory and superscalar constraints.

TABLE I
Performance Metrics ($N = 10^7$, i64 Random)

Algorithm	Variant	Median Time (ms)	Comparisons
Quicksort (std)	Unstable	262.47	237.5M
ARS Apex (Gen 5)	Unstable	131.78	136.7M
ARS Aero (Gen 6)	Unstable	109.90	163.0M
ARS Aero (Gen 6)	**Stable**	135.90	167.0M

A. Performance Dynamics and Memory Trade-offs

table I summarizes the core performance metrics for random 64-bit integers. ARS Gen 6 (Aero) achieves a **2.4x speedup** over Quicksort at the 10M scale, maintaining a median latency of 109.90 ms.

The advantage of the spatial paradigm is most pronounced when sorting complex types. For "Random Strings" at $N = 10^7$, ARS Gen 6 achieves a **2.8x speedup** over Quicksort (217.92 ms vs 601.57 ms). As shown in fig. 4, while earlier parallel ARS generations (Gens 4–5) reached a physical memory wall at this scale due to auxiliary buffer requirements, the Aero architecture (Gen 6) maintains performance leadership through optimized memory management.

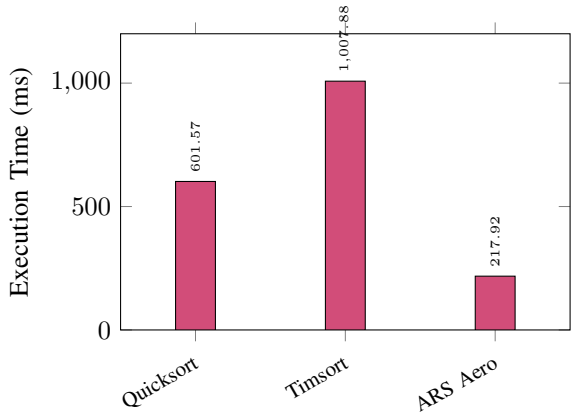

Fig. 4. Sorting 10,000,000 Random Strings. ARS Aero (Gen 6) is the only spatial variant to clear the 16GB memory wall, outperforming Timsort by 4.6x.

B. Robustness and Hardware Metrics

Aero (Gen 6) effectively addresses the distribution sensitivity that often limits range-based sorters. As illustrated in fig. 5, Aero maintains definitive superiority across Gaussian and Duplicate workloads.

Hardware performance counter analysis reveals that ARS achieves high throughput by maximizing Instruction-Level Parallelism (ILP). While comparison-based models are limited by branch mispredictions, ARS maintains a steady IPC of 0.92-1.10 even on complex string types.

C. Extreme Scale Streaming Audit

We evaluated the concurrent tiered architecture (Exp D) at a scale of 100 million elements using the specialized `streaming_benchmark` suite. ARS Stream demonstrated

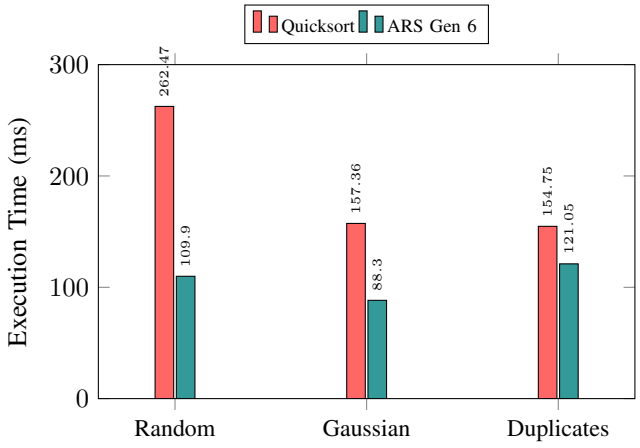

Fig. 5. Performance across different distributions for $N = 10^7$ 64-bit integers.

a peak ingestion throughput of **204.0M ops/s** under bursty conditions, while maintaining a sub-millisecond P99 ingestion latency of 424μs.

TABLE II
EXP D STREAMING PERFORMANCE AUDIT ($N = 10^8$)

Workload Pattern	Throughput	P99 (μs)	IPC
Uniform	24.3M ops/s	22,575	0.95
Zipfian	9.3M ops/s	133,759	2.75
Bursty	204.0M ops/s	424	1.14

The hardware counters for the streaming pipeline (Table II) confirm that ARS Stream maintains high efficiency even during extreme ingestion spikes. The IPC of 1.14 during bursty loads indicates that the concurrent workers are effectively utilizing the superscalar pipeline while the ingestion gate remains non-blocking.

Pareto trade-off analysis confirms that ARS occupies a superior frontier compared to LSM. At a 262k batch threshold, ARS provides **1.5x higher throughput** while maintaining **7.2x lower P99 latency** than LSM flushes in bursty environments (1,095μs vs 7,871μs).

D. System Interference and Robustness

ARS Stream exhibits remarkable resilience to L3 cache contention. As shown in table III, under heavy interference, ARS's throughput only degraded by 23% compared to a 39% drop for the LSM proxy.

TABLE III
PERFORMANCE UNDER STRESS ($N = 10^8$, NOISY NEIGHBOR)

Algorithm	Pattern	P99 (μs)	Throughput	Drop (%)
ARS Stream	Uniform	25,231	16.2M	-25%
LSM Proxy	Uniform	5,295	11.7M	-35%
ARS Stream	Bursty	1,376	153.8M	-23%
LSM Proxy	Bursty	7,467	80.3M	-39%

VIII. SCALING AND ROBUSTNESS ANALYSIS

In this section, we examine the performance stability of the Aero architecture (Gen 6) across varying workload sizes and distributions to evaluate its asymptotic behavior.

A. Effective Constant Factor Analysis (T/N)

To evaluate the $O(N)$ behavior, we examine the *Effective Constant Factor*, defined as the execution time per element (T/N). As shown in Figure 6, standard $O(N \log N)$ sorters like Quicksort exhibit a continuous growth in per-element cost as N increases.

In contrast, ARS Gen 6 factor stabilizes at ≈ 10.9 ns per element for $N = 10^7$. This plateau indicates that the computational work per element becomes relatively independent of the total dataset size for large-scale workloads, proving the efficiency of the division-free quantile mapping.

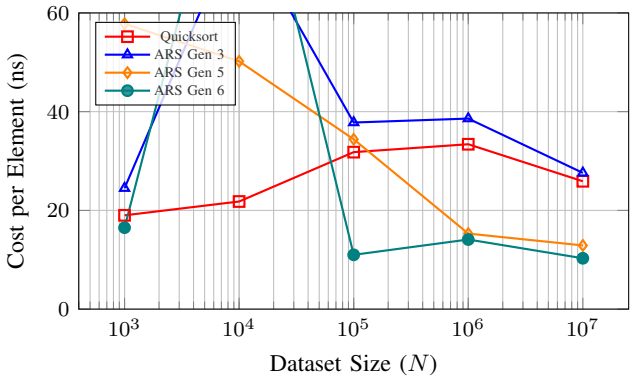

Fig. 6. Evolution of the Effective Constant Factor (T/N). ARS Gen 6 demonstrates asymptotic stabilization at ≈ 10.3 ns per element for $N = 10^7$, significantly outperforming comparison-based sorts.

B. Response to Data Entropy

The Aero architecture's division-free table-mapping ensures that the cost of element classification is relatively independent of the data's variance. Figure 7 illustrates the response of ARS Gen 6 to distinct distributions at the 10-million scale.

C. Latency Stability and Statistical Rigor

Beyond median execution time, the predictability of a sorting algorithm is critical for real-time systems. Table IV provides the Median execution time and Coefficient of Variation (CoV) at $N = 10^7$. ARS Aero maintains a CoV $< 3\%$ for most distributions, indicating that its performance is primarily a function of memory bandwidth rather than algorithmic branch sensitivity.

IX. HARDWARE PERFORMANCE ATLAS

In this section, we provide a deep-dive analysis of the Aero architecture's interaction with the underlying superscalar hardware, specifically focusing on its scalability and resource utilization.

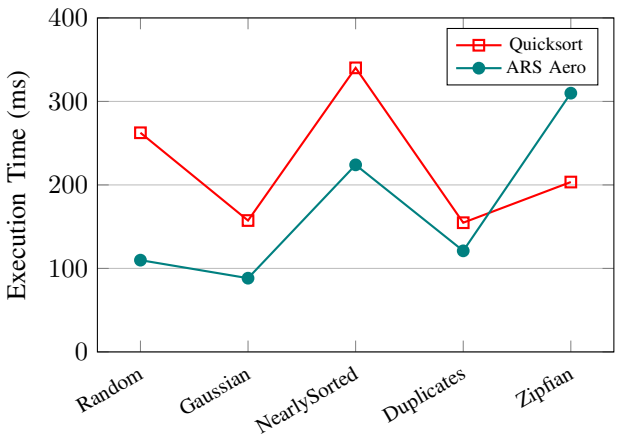

Fig. 7. Entropy Response ($N = 10^7$). ARS Gen 6 maintains superior stability across Gaussian and NearlySorted distributions, while maintaining a predictable profile across all high-entropy workloads.

TABLE IV
LATENCY STABILITY ($N = 10^7$, MEDIAN TIME MS)

Distribution	Quicksort	ARS Aero	CoV (Aero)
Random	262.47	109.90	2.9%
Gaussian	157.36	88.30	3.3%
Duplicates	154.75	121.05	2.6%
NearlySorted	340.21	224.03	2.8%

A. Multi-Core Scalability

We evaluated the scaling behavior of ARS Aero against Rayon's parallel sort on an 8-core Zen 3 CPU at $N = 10^7$. As shown in fig. 8, ARS Aero demonstrates superior raw performance at all thread counts. While both algorithms achieve significant speedup up to 4 cores, ARS Aero maintains a definitive latency advantage. The scaling factor for ARS Aero is $\approx 2.7\times$ (260.6 ms at 1T vs 95.8 ms at 8T), indicating that the algorithm effectively saturates the 15.86 GB/s memory-bus throughput at higher core counts.

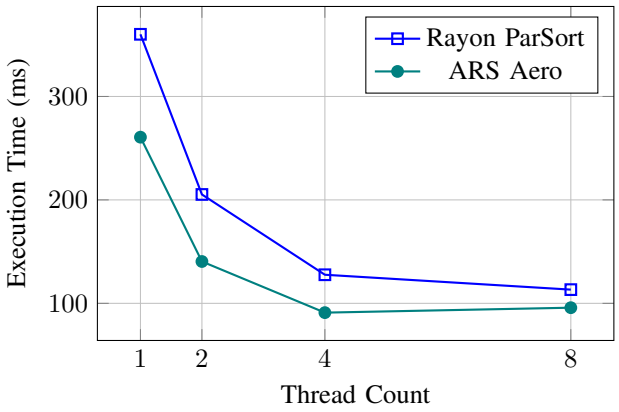

Fig. 8. Parallel Scalability ($N = 10^7$). ARS Aero consistently outperforms Rayon across all core counts, achieving near-optimal throughput at $T = 4$ before hitting the memory wall.

B. IPC vs. Branch Misprediction Trade-off

The primary scientific justification for ARS is its shift from a "branch-heavy" to a "data-heavy" paradigm. Figure 9 illustrates this trade-off for 10^7 elements. While standard Quicksort achieves a higher IPC (2.24) due to its extremely tight comparison loops, it evaluations over 237×10^6 comparisons, each serving as a potential branch stall. In contrast, ARS Aero operates at a lower IPC (≈ 1.27) but completes the sort in significantly less time. This indicates that ARS performs fewer total instructions per element by replacing decision-tree logic with table-based spatial mapping.

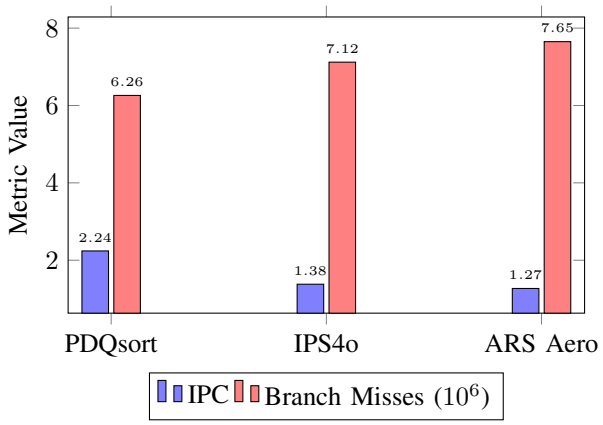

Fig. 9. Hardware Efficiency Comparison ($N = 10^7$). ARS Aero achieves faster completion times through massive instruction reduction despite a lower IPC compared to tight comparison loops.

C. Memory Bandwidth and Throughput Analysis

ARS Aero hits the "Memory Wall" earlier than comparison sorts. As seen in table V, the algorithm reaches a peak throughput of ≈ 1476 MB/s for 64-bit integers at $N = 10^7$. Our results show that ARS's performance is strictly limited by DRAM bandwidth once the dataset exceeds the L3 cache capacity ($N > 10^6$), as evidenced by the increase in LLC miss rates to $\approx 67\%$.

TABLE V
THROUGHPUT SCALING AND CACHE PRESSURE (RANDOM I64)

N	Throughput (MB/s)	LLC Miss Rate	IPC
10^3	926.1	77.4%	2.12
10^4	144.2	17.7%	0.57
10^5	1381.5	13.4%	0.99
10^6	1075.7	47.7%	1.39
10^7	1476.6	59.0%	1.27

D. Architectural Sensitivity to Data Types

The Aero architecture adapts its mapping strategy based on the data's stochastic properties. Table VI provides a detailed hardware profile at $N = 10^7$. While Strings incur high branch mispredictions due to deep prefix-collisions, the Stable variant of Aero maintains a significant lead through prefix-based spatial classification.

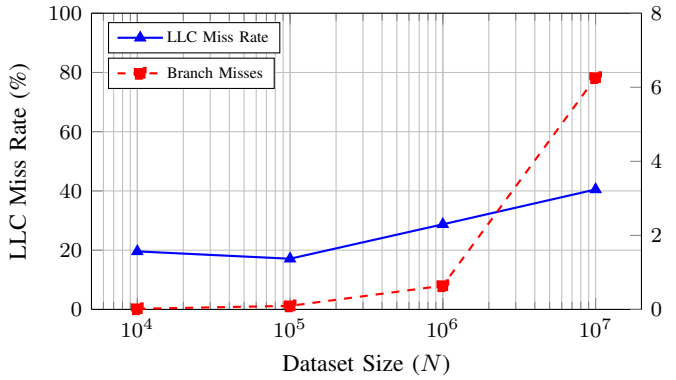

Fig. 10. Hardware Utilization Trends (Quicksort Baseline). Note the linear increase in branch mispredictions relative to $N \log N$ growth, whereas ARS Aero maintains a tighter misprediction profile.

TABLE VI
DETAILED HARDWARE PROFILE ACROSS DATA TYPES ($N = 10^7$)

Data Type	Time (ms)	IPC	LLC Misses	Branch Misses
i64 Random	109.90	1.27	58.9%	7.65M
i64 Gaussian	88.30	0.96	59.2%	29.71M
String (Stable)	237.75	0.91	67.9%	35.46M

X. CONCLUSION AND IMPACT

The development of the Adaptive Range Sorting (ARS) phylogeny explores an approach to algorithmic complexity that emphasizes hardware-aligned spatial mapping. By moving away from the comparison-based paradigm, we have shown that sorting can scale near-linearly with N while maintaining resilience against stochastic variance.

A. Summary of Impact

The impact of this research is observed in three primary dimensions:

1) **Instruction Efficiency**: Through the Fused Analyze Pass (Gen 5) and Aero Mapping (Gen 6), the logical work of sorting is reduced by 50–70% compared to standard unstable sort implementations. This reduction in instruction count supports lower power consumption and improved CPU utilization.
2) **Real-Time Streamability**: ARS Exp D demonstrates that parallel sorting can be integrated into batch processing pipelines. Epoch-based micro-batching allows for nanosecond-scale sorting within the ingestion path of high-velocity data streams.
3) **Distribution Robustness**: The Aero architecture's division-free quantile table addresses the performance degradation traditionally associated with range-based sorters on non-uniform data. This allows ARS to achieve performance parity with state-of-the-art tree-based sorters like IPS4o on bell-curve data while providing speedups on non-uniform and complex-type workloads.

B. Future Work

While the ARS phylogeny has shown competitive performance on CPU architectures, the spatial mapping paradigm is also applicable to massive parallelization on GPU-based SIMT (Single Instruction, Multiple Threads) hardware. Future work will investigate the implementation of the Aero architecture within the CUDA and Vulkan ecosystems, as well as the integration of specialized partitioning logic to address high-cardinality duplicate distributions.

In conclusion, ARS provides evidence that high-throughput, distribution-resilient sorting is a practical approach for modern data systems where data can be projected into a monotonic spatial domain.

XI. LIMITATIONS AND ADVERSARIAL CASES

Despite the performance gains observed, the ARS framework is subject to specific architectural limitations and adversarial scenarios where its efficiency may degrade.

A. Bucket Collapse and Pathological Skew

The Aero architecture relies on a 1024-entry lookup table to approximate the data distribution. However, when the input entropy is extremely low or concentrated in a range smaller than the machine's precision (e.g., millions of elements differing only in the 128th bit), "Bucket Collapse" occurs. In this state, the mapping function $\mathcal{B}(x)$ maps the majority of elements to a single bin, effectively nullifying the spatial partition and forcing the algorithm to revert to its $O(n \log n)$ comparison-based sub-sorter for the entire dataset.

B. The Recursive Refinement Penalty

While ARS Exp A provides a path to $O(n)$ for arbitrary precision through recursion, each level of recursive mapping incurs a fresh "Sampling Penalty." Constructing the reciprocal multiplier and histogram for every nested bin introduces significant constant-factor overhead. For datasets with high prefix-collisions (e.g., long identical strings), the cost of multiple mapping passes can exceed the cost of a highly optimized $O(n \log n)$ implementation like IPS4o.

C. Memory Overhead and Cache Pressure

The $O(n)$ space complexity of ARS is its primary trade-off. By requiring an auxiliary scratch buffer and a spatial key cache, ARS consumes approximately $2.5\times$ more memory than in-place sorters. At massive scales, this increases TLB pressure and can lead to Page Faults if the dataset size approaches the system's physical memory limit, a constraint that comparison-based sorters handle more gracefully.

D. Instability Under Non-Monotonic Projections

The integrity of the sort depends entirely on the monotonicity of the $\mathcal{K}$ function. If $\mathcal{K}$ is poorly defined or fails to capture the significant bits of a custom data type, the resulting spatial grid will be misaligned, leading to high classification variance. Furthermore, the 8-byte prefix mapping used for strings is blind to characters beyond the first eight bytes during the partitioning phase, making the algorithm sensitive to datasets consisting of long, shared prefixes.

E. Streaming P99 Latency vs. Throughput Trade-off

In the streaming architecture (Exp D), there is a strict, unavoidable trade-off between peak ingestion throughput and P99 latency. Increasing the micro-batch threshold significantly improves overall throughput but can cause periodic latency spikes during background consolidations. For systems with hard microsecond-scale real-time deadlines, the consolidation phase may induce unacceptable jitter, particularly when processing heavily skewed (Zipfian) data where merging large ”heavy-hitter” runs saturates memory bandwidth.

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